UNIT QUESTIONS

SKILLS PROBLEM SOLVING 7

1 **a** Which of the following particles has the smallest mass?

A alpha

B neutron

C proton

D electron

(1)

SKILLS CRITICAL THINKING 8

b When a radioactive isotope emits an alpha particle its

A atomic mass decreases and its atomic number decreases

B atomic mass increases and its atomic number decreases

C atomic mass decreases and its atomic number increases

D atomic mass is unchanged and its atomic number decreases

(1)

c The emission of which of the following particles from a radioisotope causes it to change into the next element in the periodic table?

A positron

B electron

C neutron

D alpha

(1)

SKILLS PROBLEM SOLVING 9

d A radioactive isotope has an activity of 1000 Bq; after 5 hours its activity has fallen to 125 Bq. The half-life of this isotope is

A 1 hour

B 75 minutes

C 125 minutes

D 2.5 hours

(1)

SKILLS CRITICAL THINKING 8

e Which of the following particles that may be emitted in radioactive decay is not ionising?

A alpha

B beta minus

C neutron

D beta plus

(1)

(Total for Question 1 = 5 marks)

SKILLS PROBLEM SOLVING

2 In the Sun and other stars the temperature and pressure make hydrogen nuclei fuse together to form heavier nuclei. This process produces a very large amount of energy, various types of radiation and helium. The steps of this fusion process are shown in the following nuclear equations.

Balance these equations by filling in the boxes:

a $^{1}_{1}H + {}^{1}_{1}H \rightarrow {}^{\square}_{1}H + {}^{0}_{\square}e$ **(2)**

b $^{2}_{1}H + {}^{1}_{1}H \rightarrow {}^{\square}_{\square}He$ **(2)**

c $^{\square}_{2}He + {}^{\square}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{\square}\square + {}^{1}_{\square}\square$ **(3)**

(Total for Question 2 = 7 marks)

3 Here are descriptions of some nuclear particles:

Particle A has 0 mass and a charge +1.

Particle B has 1 mass and a charge 0.

Particle C has 1 mass and a charge +1.

Particle D has 0 mass and a charge –1.

Particle E has 4 mass and a charge +2.

a State which particle is an alpha. **(1)**

b State which particle is a beta minus. **(1)**

c State which particle is a positron. **(1)**

d State which particle is a proton. **(1)**

e State which particle is a neutron. **(1)**

(Total for Question 3 = 5 marks)

SKILLS CRITICAL THINKING

4 Two types of nuclear reactor are used to produce energy. Fission reactors are already in use in nuclear power plants around the world. Fusion reactors are still at the experimental stage.

a Explain what is meant by:

i nuclear fission **(4)**

ii nuclear fusion **(4)**

b In a fission reactor the process of generating energy is a chain reaction using a suitable fissile fuel like uranium-235.

i Explain what a chain reaction is, and **(4)**

ii how it be controlled or shut down completely. **(4)**

(Total for Question 4 = 16 marks)

SKILLS CRITICAL THINKING 9 SKILLS REASONING 8

5 **a** Explain the conditions that are needed to cause nuclear fusion. **(3)**

b State two reasons why nuclear fusion reactors may be a better alternative to nuclear fission reactors. **(2)**

(Total for Question 5 = 5 marks)

SKILLS CRITICAL THINKING 7 SKILLS PROBLEM SOLVING 9

6 **a** Explain what the letters A and Z tell us about the structure (make up) of the nucleus of any element X.

$^{A}_{Z}X$ **(2)**

b Here is a list of the type of radiation that may be emitted (given out) by an unstable nucleus:

A alpha **B** beta **C** gamma **D** neutron

i Which type affects A but not Z? **(1)**

ii Which type affects Z but not A? **(1)**

iii Which type affects both A and Z? **(1)**

iv Which type affects neither A nor Z? **(1)**

(Total for Question 6 = 6 marks)